COURSE PROJECT · STATISTICAL METHODS IN RELIABILITY (RE60013)

Downtime Analysis & Repair Time Prediction in Manufacturing

Statistical analysis of 635 stoppage events on a heavy-clay production line

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THE PROBLEM

One stopped machine halts the whole line

A serial production line with minimal buffers means every repair minute becomes lost output. Repair times vary widely, and management has six questions.

01

Is the average repair time below target?

02

Are repair times too variable?

03

Did the new maintenance procedure help?

04

Which factors drive long repairs?

05

Are long repairs acceptably rare?

06

Can we predict a future repair time?

DATASET & ASSUMPTIONS

Real MES data from a heavy-clay brick line

635
stops before Lean Six Sigma

540
stops after Lean Six Sigma

10,388 min
total downtime before LSS

The brief asks for	Our dataset provides
Repair time	Stop duration (minutes)
New maintenance procedure	Before vs after Lean Six Sigma
Machine type	Stop location
Failure severity	Stop group (used as proxy)
Operating shift	Derived from stop start time
Machine age, technician experience	Not recorded (latent variables)

Source: Ntoas (2025), OEE and Downtime Dataset, Zenodo 17855209 · Evocon OEE software

PART A · Q1–Q3

Defining repair time and what drives it

Q1

Response

- Starts at stop start time
- Ends at stop end time
- Measured in minutes
- Includes spare-part waits (tagged MATERIALS)

Q2

Factors

- Machine / stop location
- Stop reason and group
- Shift (from start time)
- Product code (load)
- Lean Six Sigma period

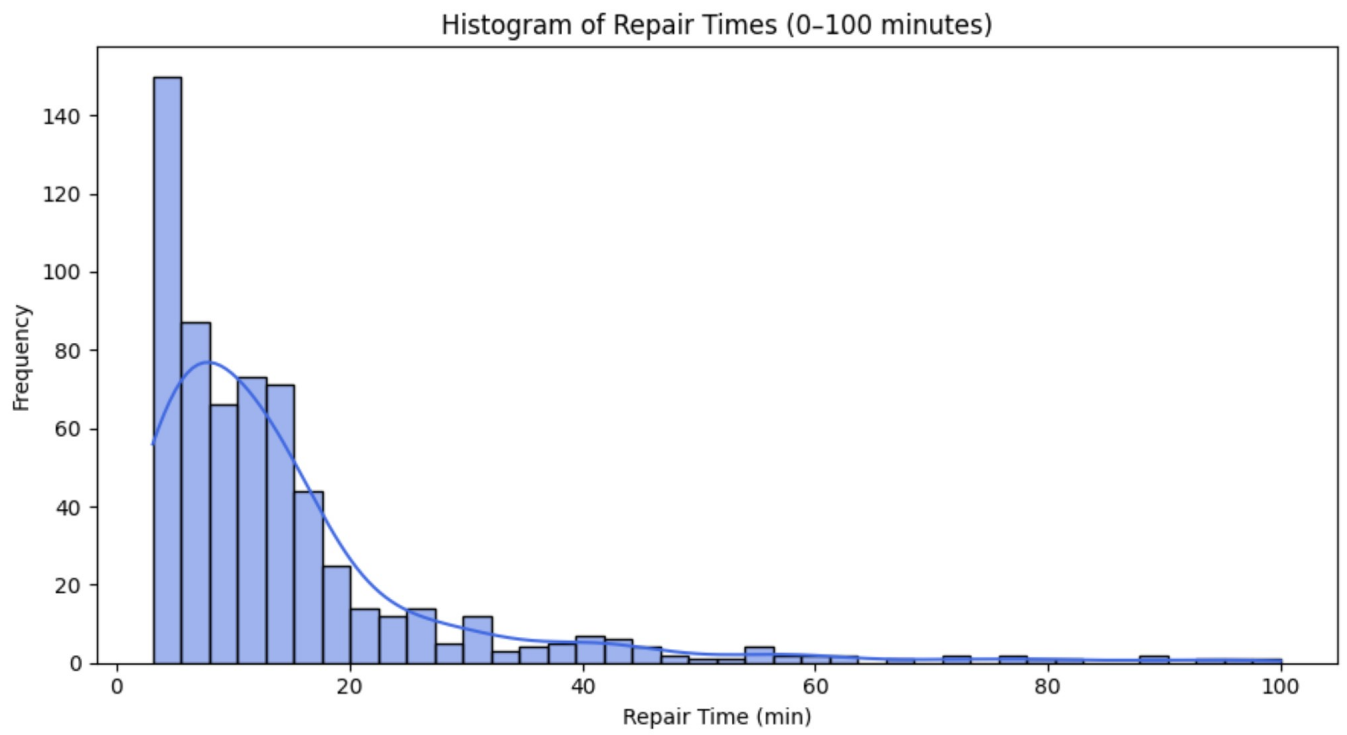
Q3

Data quality

- Full MES log, no sub-sampling
- Bias: no technician or machine-age data
- Dropped empty columns
- Blank notes imputed, types fixed

PART B · Q4 · DISTRIBUTION

Repair times are heavily right-skewed

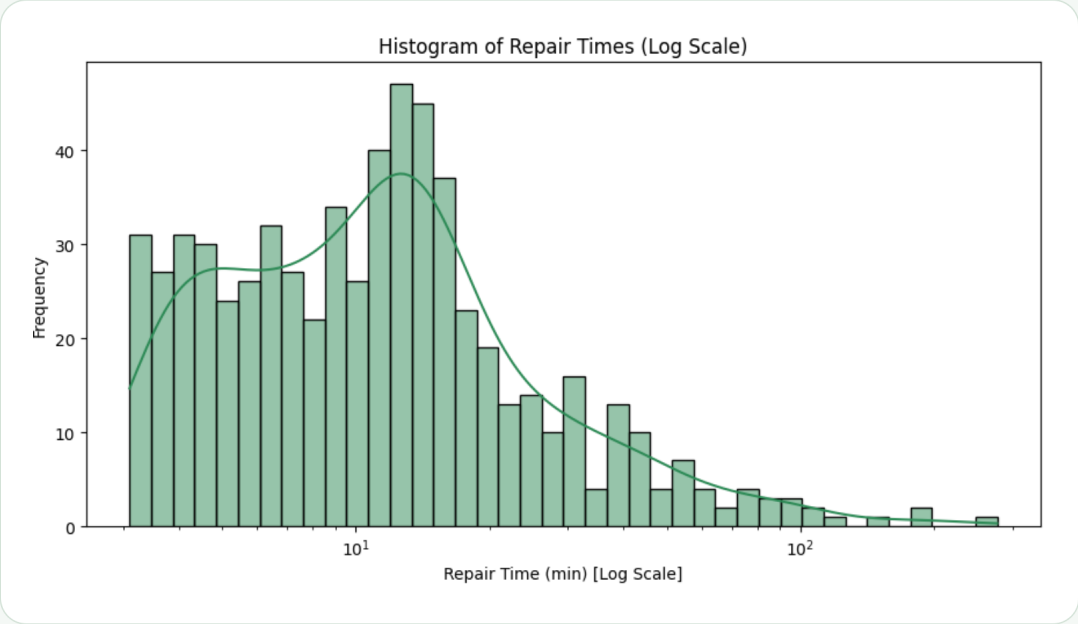


Mean (min)	16.36
Median (min)	11.00
Std. deviation (min)	22.19
Range (min)	273.70
Skewness	5.60
Kurtosis	45.72

Mean above median: a few extreme repairs pull the average up.

PART B · Q5–Q6 · PROBABILITY MODEL

Lognormal: the best model for repair time



On a log scale the data is roughly bell-shaped. Fitted: σ = 0.811, scale = 10.97 min, location = 0.

Q5 · CANDIDATE MODELS

Model	Fit	Reason
Normal	No	Symmetric
Exponential	No	Peaks at zero, one parameter
Gamma/Weibull	Possible	Skewed, but lighter tail
Lognormal	Chosen	Positive, heavy right tail

Q6 · PROBABILITIES FROM THE MODEL

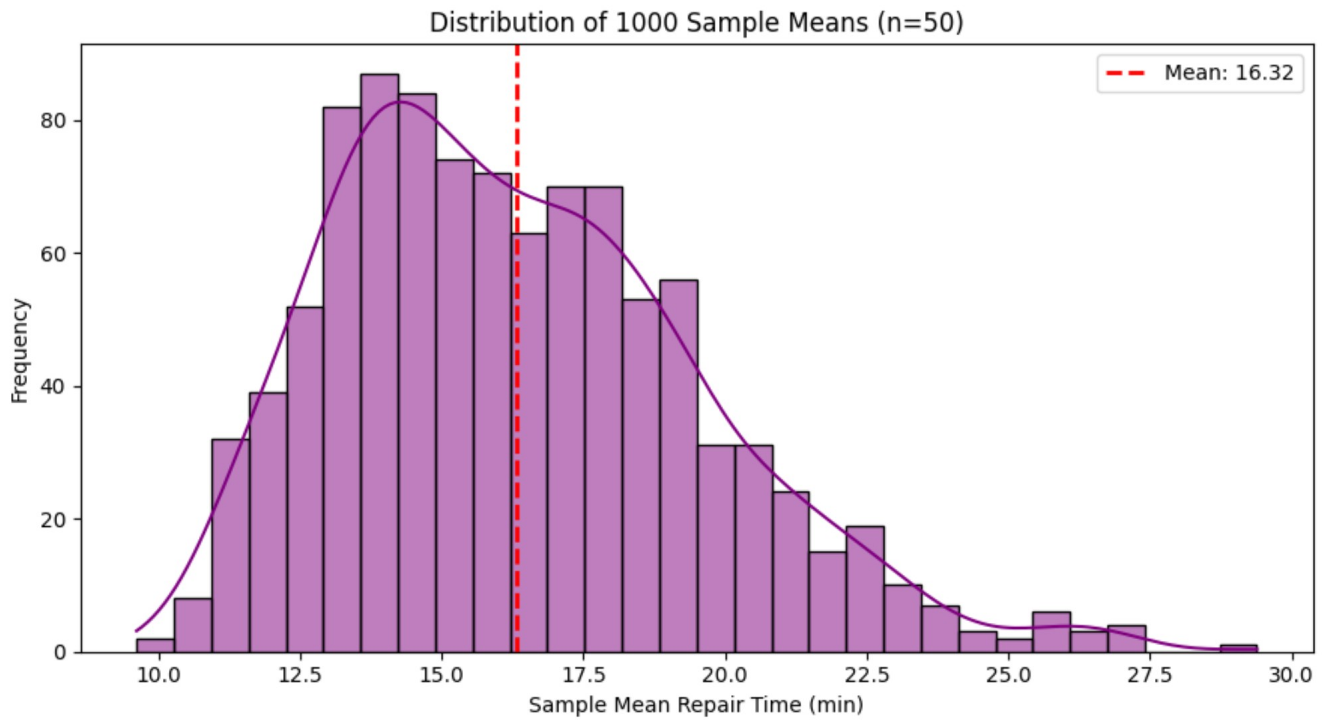
1.8%
P(repair > 60 min)

89.3%
P(repair ≤ 30 min)

65.0%
P(repair < 15 min)

PART C · Q7 · SAMPLING VARIABILITY

Another sample gives a different mean



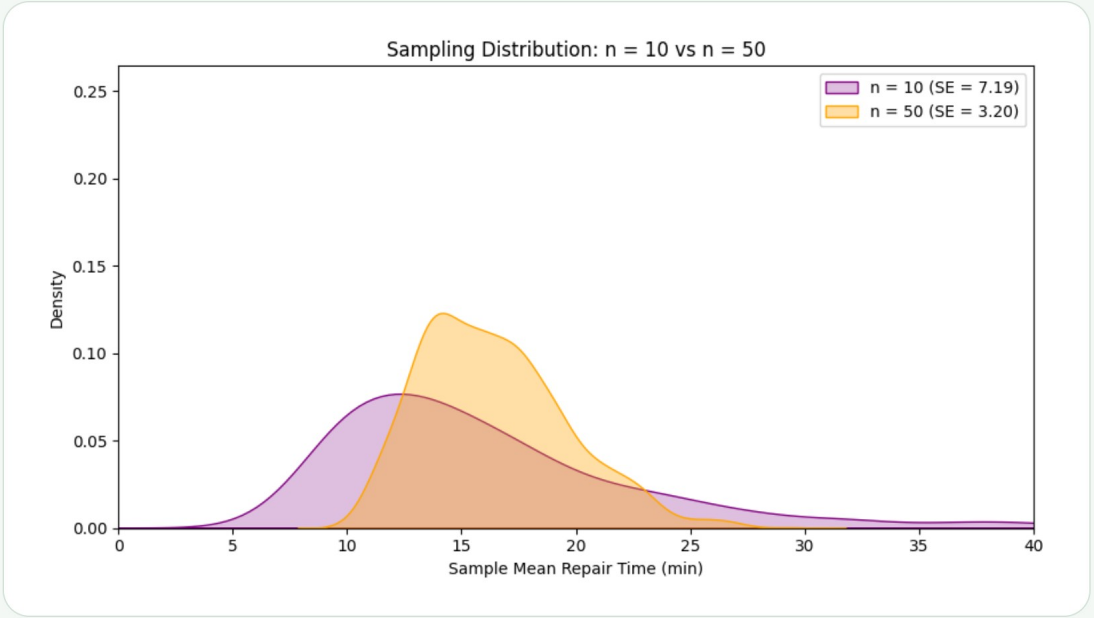
1,000 SAMPLES OF N = 50

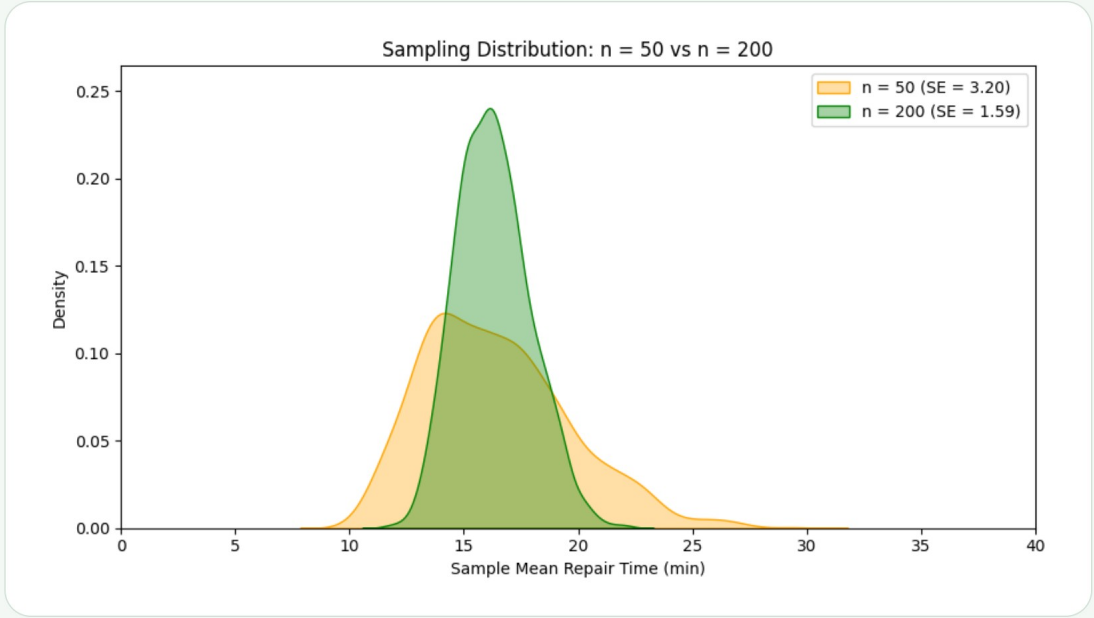
Lowest mean (min)	9.62
Highest mean (min)	29.39
Mean of means (min)	16.32
Standard error (min)	3.23

No. The mean shifts from sample to sample, by up to 20 minutes at n = 50.

PART C · Q8 · SAMPLE SIZE

Larger samples give sharper estimates





SE at n = 10
7.19 min

→

SE at n = 50
3.20 min

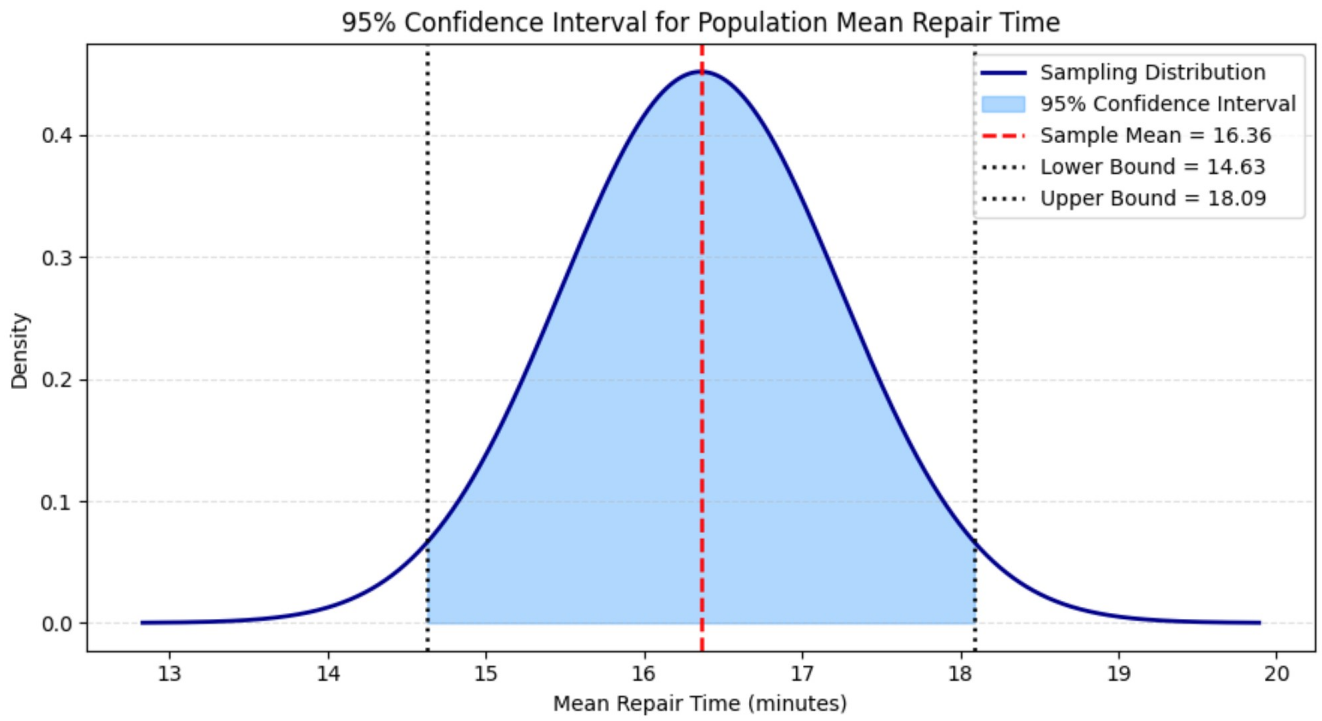
→

SE at n = 200
1.59 min

4× the sample ≈ half the error, as σ/√n predicts.

PART D · Q9–Q10 · ESTIMATION

The true mean lies between 14.6 and 18.1 min



Best point estimate (Q9)

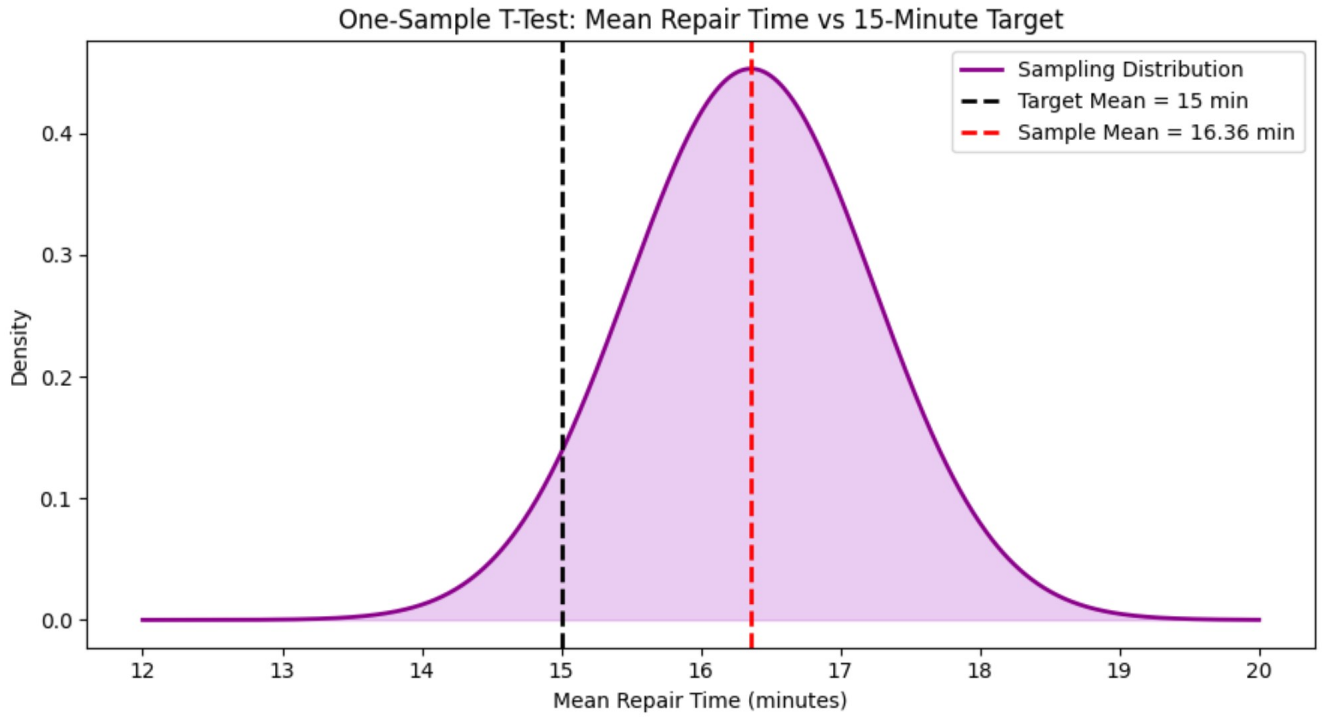
16.36 min

95% CI lower (min)	**14.63**
95% CI upper (min)	**18.09**
Sample size	**635**

95 of 100 intervals built this way would contain the true mean.

PART E · Q11 · ONE-SAMPLE T-TEST

Mean repair time vs the 15-minute target



H_0: μ = 15 · H_1: μ > 15

Sample mean (min)	**16.36**
t-statistic	**1.54**
p-value	**0.0616**

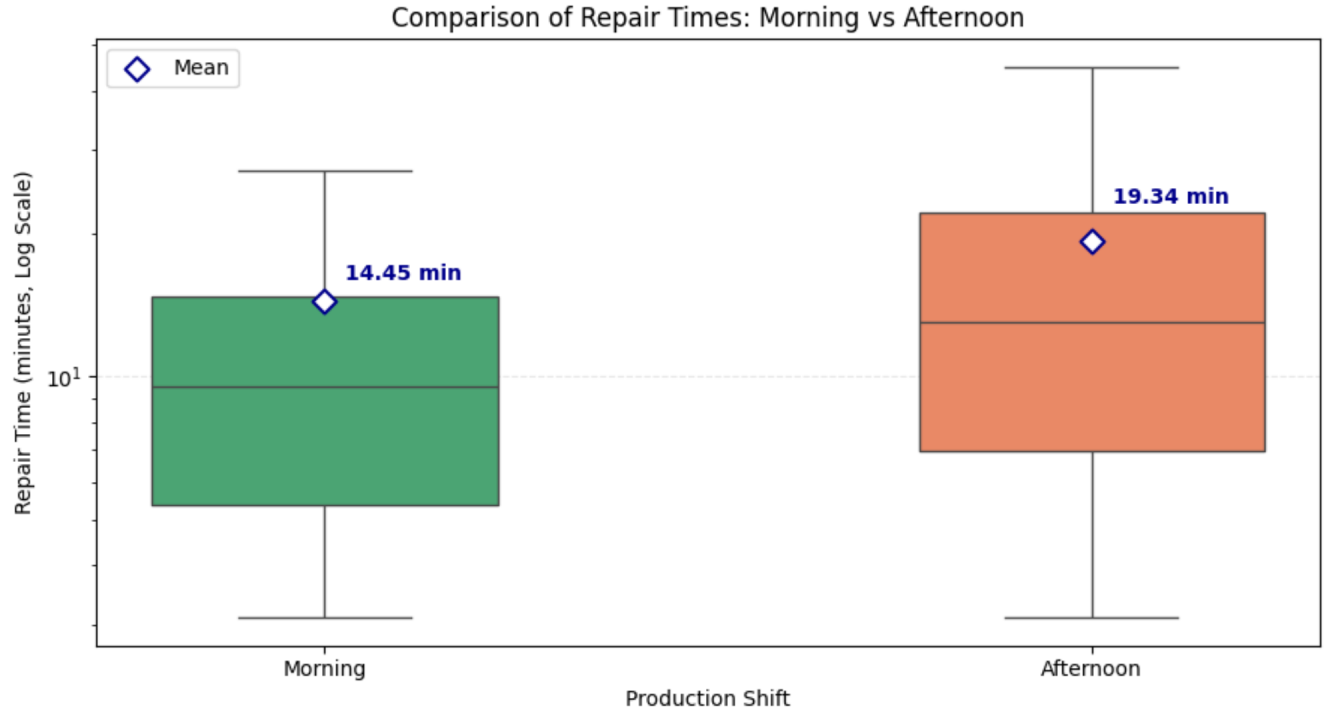
Fail to reject H_0

Above 15 min, but not significantly. No evidence the target is breached.

Target adjusted from 60 to 15 min to suit the data.

PART E · Q12 · WELCH TWO-SAMPLE T-TEST

Afternoon repairs take about 5 minutes longer



H_0: μ morning = μ afternoon

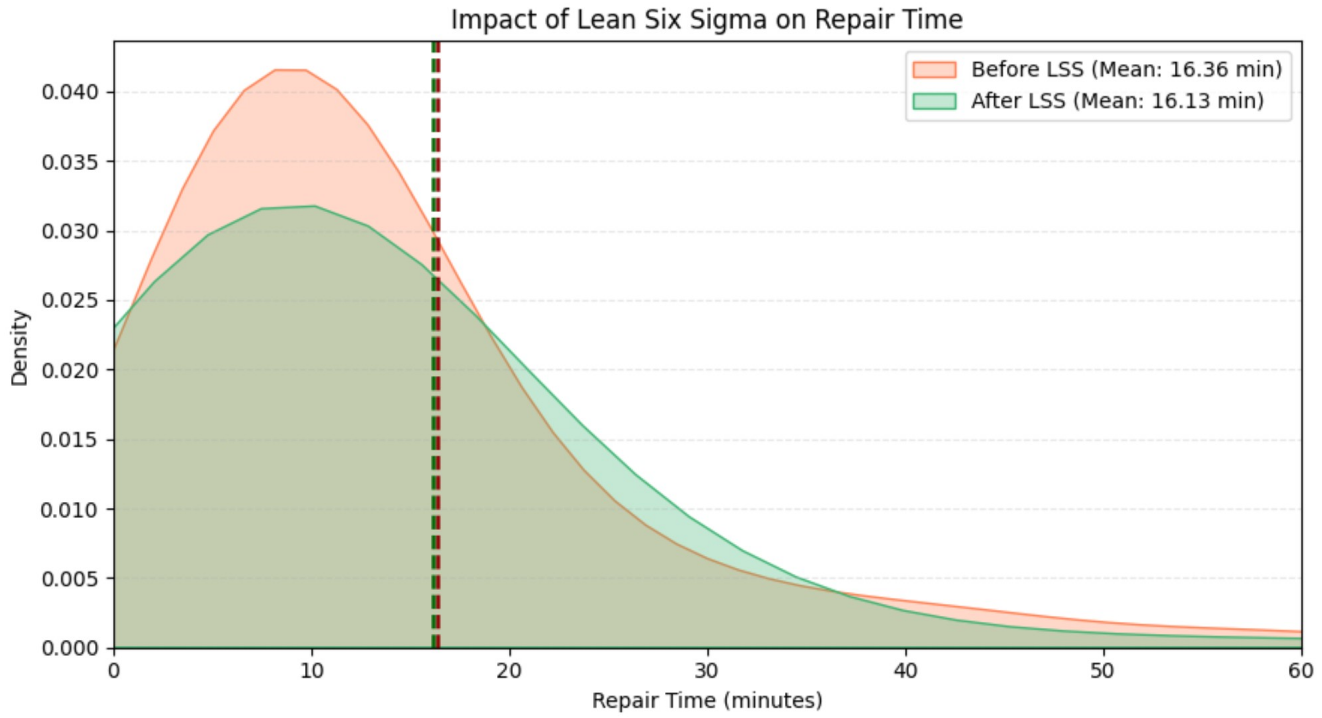
Morning (n = 387)	14.45
Afternoon (n = 248)	19.34
t-statistic	−2.81
p-value	0.0051

Reject H_0: significant

Means in minutes. Day/night adapted to morning/afternoon, per the timestamps.

PART E · Q13 · TWO-SAMPLE T-TEST

Lean Six Sigma did not cut repair time



H_0: μ before = μ after · H_1: μ before > μ after

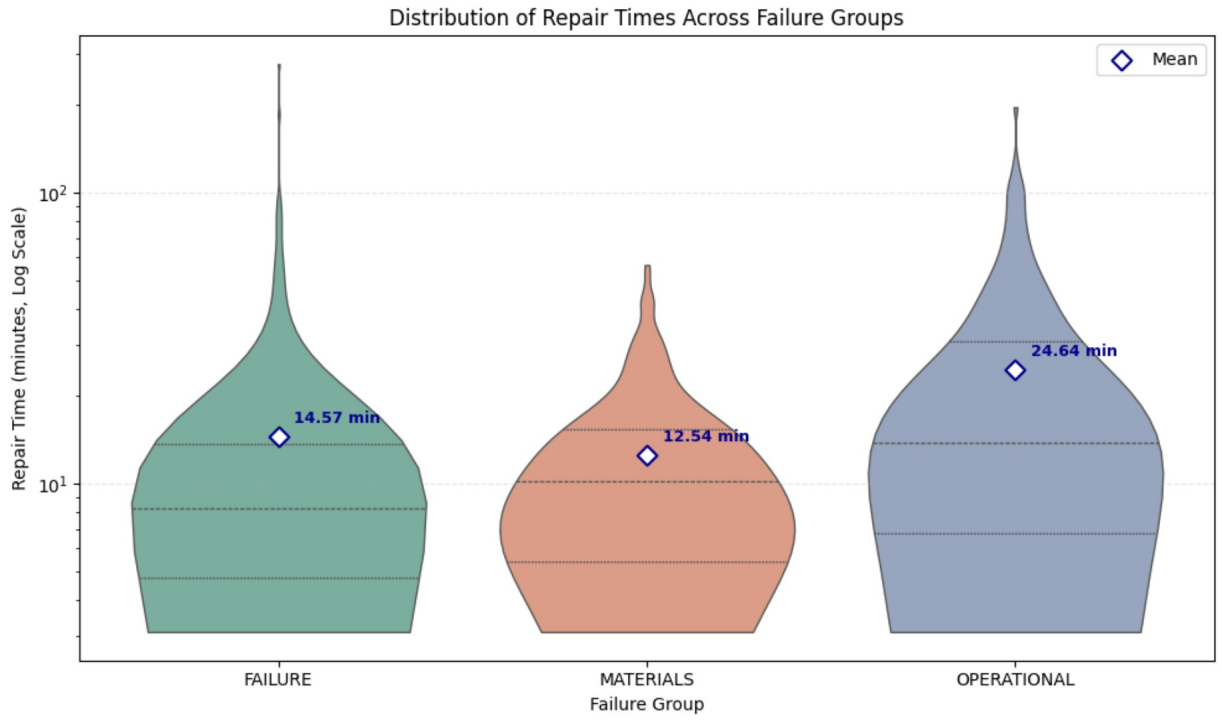
Before LSS (n = 635)	**16.36**
After LSS (n = 540)	**16.13**
t-statistic	**0.13**
p-value	**0.4475**

Fail to reject H_0

A 14-second drop, indistinguishable from noise.

PART E · Q14 · ONE-WAY ANOVA

Operational stops take the longest to fix



H_0: all group means are equal

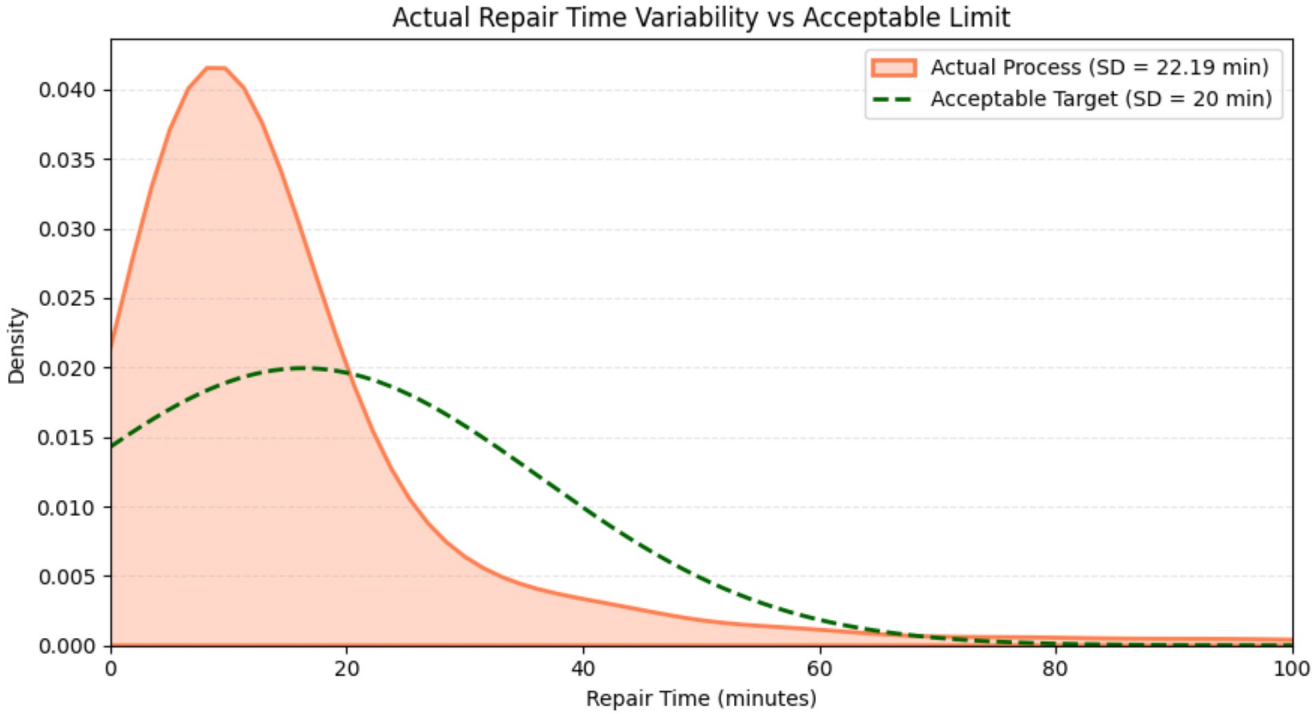
Failure (n = 222)	14.57
Materials (n = 189)	12.54
Operational (n = 127)	24.64
p-value	8.45×10^{-6}

Reject H_0: significant

Means in minutes. Stop group used as a proxy for severity.

PART E · Q15 · CHI-SQUARE TEST FOR VARIANCE

Repair-time variability exceeds the limit



H_0: $\sigma^2 = 400$ · H_1: $\sigma^2 > 400$

Limit (SD 20 min)	400
Sample (SD 22.19 min)	492.24
χ^2 statistic	780.19
p-value	6.03×10^{-5}

Reject H_0: too variable

The spread, not the average, is the problem.

PART E · Q16 · EXACT BINOMIAL TEST AT THREE THRESHOLDS

More than 10% of repairs exceed 25 minutes

THRESHOLD	OBSERVED	P-VALUE	DECISION
> 25 min	14.33%	0.0003	Reject H_0
> 30 min	11.18%	0.1765	Fail to reject H_0
> 35 min	8.98%	0.8222	Fail to reject H_0

What we tested

H_0: $p \leq 0.10$ vs H_1: $p > 0.10$, at $\alpha = 0.05$. Is the share of repairs over each threshold above 10%?

What it means

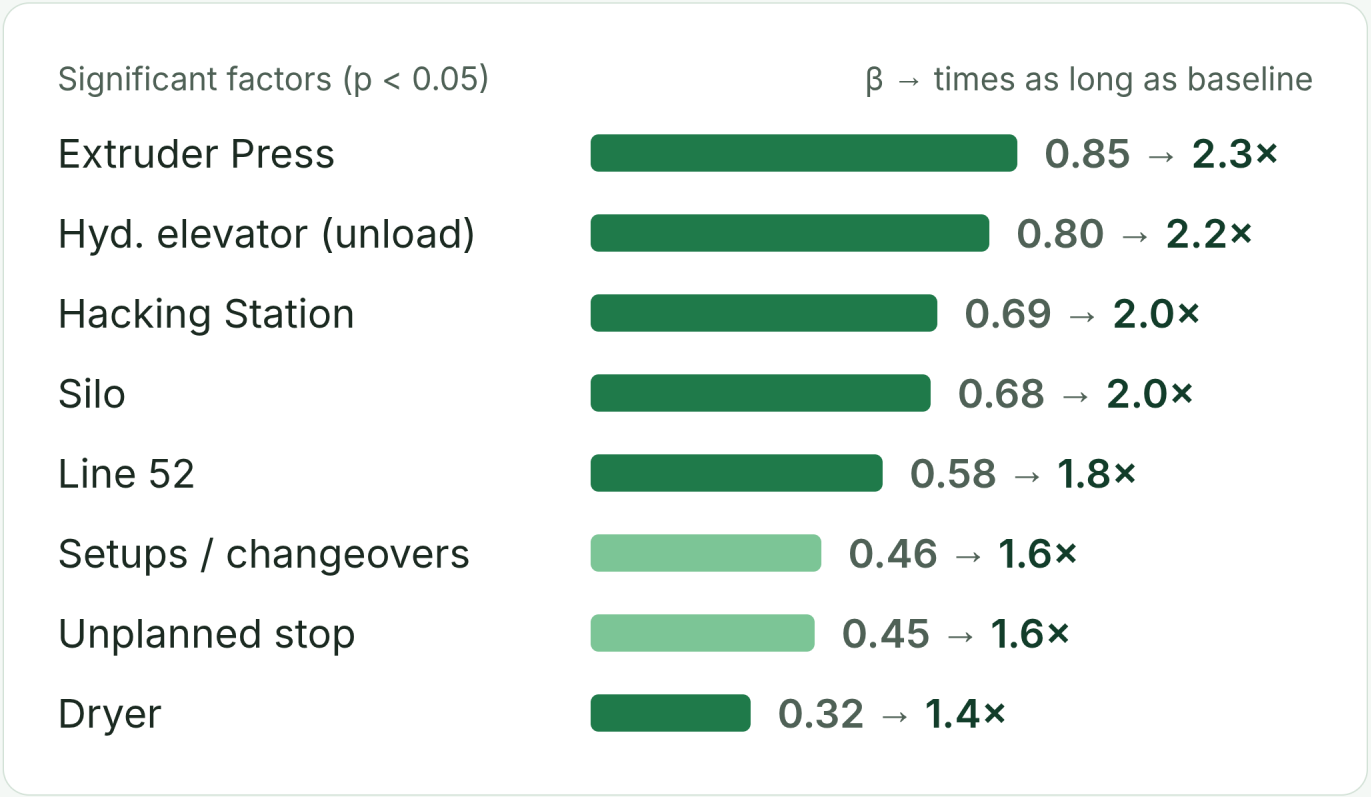
Too many repairs run past 25 min. Past 30 or 35 min, the share is not significantly above 10%.

Engineering takeaway

Investigate repairs over 25 min for causes of delay. The tests share data: read them as a sensitivity check.

PART F · Q18 · MULTIPLE LINEAR REGRESSION

Where a machine stops matters most



1. Regression measures each factor's effect on repair time while holding the other factors fixed.
2. Repair times are skewed, so we model log(repair time). Each β then acts as a multiplier: time × e^β.
3. β compares a category with a baseline (e.g. the Cutter). β = 0.69 gives e^0.69 ≈ 2: twice as long.

R^2: model explains 22% of the variation **0.219**

PART F · Q19 · PREDICTION

Predicting a future Extruder Press stop

EXAMPLE STOP **Machine:** Extruder Press **Group:** Operational **Type:** Unplanned **Starts:** 14:00 (afternoon)

Predicted median repair

18.7 min

95% prediction interval: 4.2 to 83.1 min



1 · How we predict

We feed this stop's details into the slide 16 regression. It gives a log(time), which we convert back to minutes.

2 · Why the median

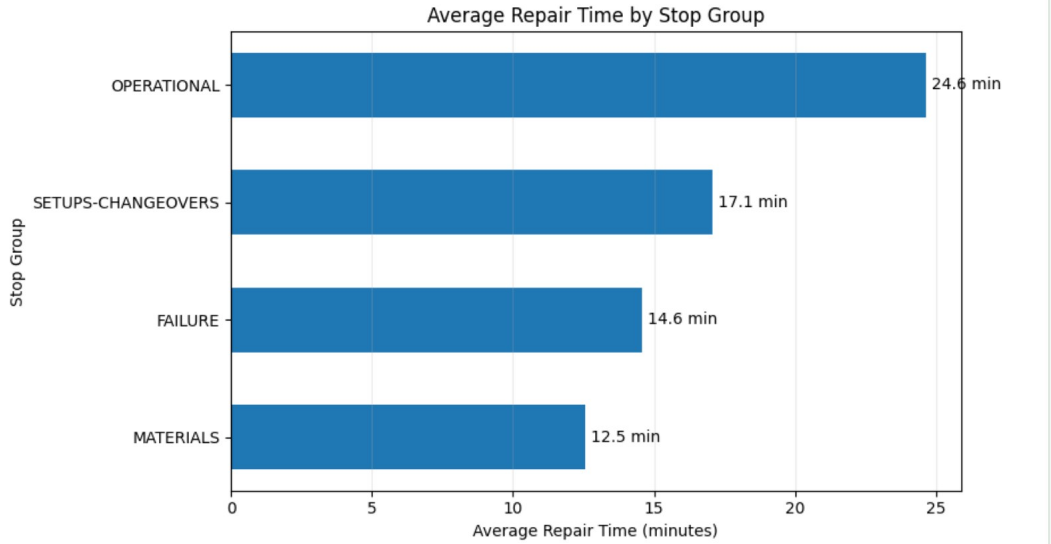
Converting back from logs gives the median: half of such repairs end before 18.7 min, half after.

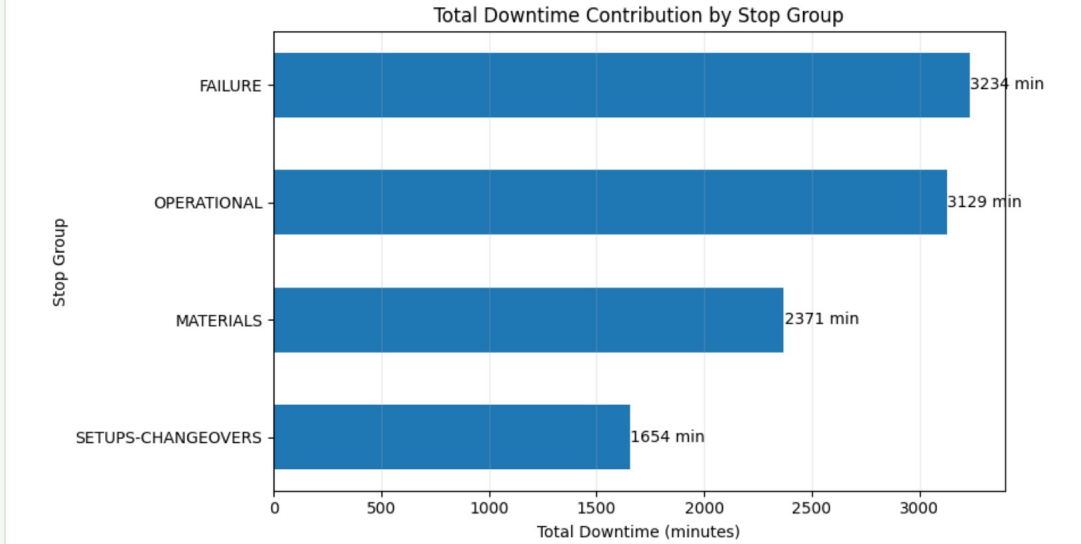
3 · Why the range is wide

It covers one single repair, not the average, so it can exceed 80 min. Missing data (severity, skill) widens it.

PART G · WHERE DOWNTIME COMES FROM

Two machines cause over half of all downtime





29.3%
Extruder Press downtime

24.7%
Hacking Station downtime

31.1%
from mechanical failures

59.5%
of downtime is unplanned

PART G · Q20 · RECOMMENDATIONS

Four ways to cut downtime

WHAT TO DO	WHAT WE FOUND	HOW TO DO IT
01 Fix the two slowest machines first	The Extruder Press and the Hacking Station cause more than half of all downtime.	Keep spare parts and tools right next to them, and service them before they break.
02 Let operators clear simple jams	Jams and other process problems take the longest to fix, about 25 minutes each.	Train machine operators to clear them on the spot, instead of waiting for a mechanic.
03 Support the afternoon shift	Repairs in the afternoon take about 5 minutes longer than in the morning.	Make sure the afternoon has enough staff, experienced technicians, and easy access to parts.
04 Act fast on long repairs	The new LSS method did not speed up repairs, and a few repairs still drag on for hours.	If a machine is not fixed within 30 minutes, call a senior engineer. Study the slowest repairs to learn why.

CONCLUSION

What the evidence says

Question	Evidence	Verdict
Average repair time	16.36 min vs 15-min target, $p = 0.062$	Within target
Variability	SD 22.19 vs 20-min limit, $p < 0.001$	Too high
New procedure (LSS)	16.36 → 16.13 min, $p = 0.45$	No effect
Trend over time	−0.75 min per month, $p = 0.51$	No change
Drivers	Location, stop group, stop type, shift	Identified
Long repairs	14.3% over 25 min vs 10% limit, $p < 0.001$	Too many at 25 min
Prediction	Median 18.7 min, 95% PI 4–83 min	Possible, wide

Thank you